

Dekun Yang

✉ dy305@cam.ac.uk | 📞 +44 07733466857

EDUCATION

University of Cambridge | *Natural Science(Physics) B.A. and M.Sci.* **2022-present**

- B.A. result: Class II.1 (GPA 3.5+)
- Related Coursework: Advanced Quantum Physics, Quantum Condensed Matter Physics, Statistical Mechanics, Electrodynamics and Optics, Soft Condensed Matter Physics, Python Programming, Scientific Writing

Concord College | *A-level* **2020-2022**

- Grade: A*A*A*A (Further Math, Math, Physics, Chemistry)
- Plus Extended Project Qualification focused on fluid dynamics

RESEARCH INTERESTS

Application of classic and advanced computational approaches, including numerical simulations, machine learning, and deep learning, to study problems in condensed matter physics and materials science/engineering.

PUBLICATION AND WORKING PAPER

Dylan Shearsby, Jiaqi Wu, **Dekun Yang**, Bo Peng*, Tuning electronic and optical properties of 2D polymeric C_{60} by stacking two layers. Published on *Nanoscale*, 2025, 17, 2616-2620. <https://doi.org/10.1039/D4NR04540H>

Dekun Yang, Cameron A.M. Scott, Dylan Shearsby, Nicholas Bristowe, Daniel Bennett*, Bo Peng*, Structural stability and Symmetry analysis of novel bismuth oxide ferroelectric. In preparation.

Dekun Yang, Daniel Bennett and Bo Peng*, Novel ferroelectrics based on polar molecules encapsulated inside fullerenes. In preparation.

RESEARCH EXPERIENCE

Pennsylvania State University | *Research Intern* **July 2025 - Present**
Supervised by Prof. Binghai Yan, Department of Physics

Project Topic:

Examining Charge Density Wave(CDW) and possible superconductivity in novel Monolayer material

- Identified a CDW-perturbed structure by applying symmetry analysis and targeted structural perturbations
- Generated simulated Scanning Tunneling Microscopy (STM) images, observing a characteristic contrast inversion in the periodic charge distribution that successfully matched experimental results
- Surveying literature and consulting prior collaborators to identify tractable ways to explore exciton properties in large supercell

Shanghai Jiao Tong University | *Visiting student* **August 2025 - October 2025**
Supervised by Prof. Hengxin Tan, Institute of Condensed Matter Physics

Project Topic:

Examining Charge Density Wave(CDW) and possible superconductivity in novel Monolayer material

University of Cambridge | Research Intern **December 2024 - Present**
Supervised by Dr. Bo Peng, Theory of Condensed Matter Group (UoC)

Project-1:

Working with Dr. Ruoshi Jiang, Theory of Condensed Matter Group (Univ. of Cambridge)

Unconventional superconductivity in layered fullerene structures

- Investigated evidence of strong electron correlation within the target structure
- Built and validated a tight-binding model treating the C60 buckyball as a carbon super-atom, enabling further calculations and analysis of the system

Project-2:

Design of novel 2D ferroelectric materials using buckyballs

- Proposed possible chemical compositions and crystal structures based on previous theoretical assumptions
- Identified a few compatible FE candidates with reasonable polarization by performing first-principles calculations using VASP, Wannier90, and Quantum Espresso.
- Determined that charge transfer between central ions and the buckyball cage was the mechanism responsible for reduced polarization in other candidate structures.

University of Cambridge | Summer Research

Supervised by Dr. Bo Peng, Theory of Condensed Matter Group (UoC)

Project-1:

June 2024 – October 2024

Tuning electronic and optical properties of 2D polymeric C_{60} by stacking two layers.

- Contributed to DFT calculations for electronic band structure and photo-absorption properties
- Found stronger exciton absorption in the bilayer than that in the monolayer over the entire visible light range, proposing the bilayer material as a more promising candidate for photovoltaic applications

Project-2:

June 2024 – Present

Co-advised by Prof. Daniel Bennett (Nanyang Technological University, Harvard University). Working with Dr. Cameron A.M. Scott and Prof. Nicholas Bristowe (Durham University).

Structural stability and Symmetry analysis of novel bismuth oxide ferroelectric

- Determined the stable bulk structure of Bismuth Oxide (Bi_6O_9) by combining a group-theory-based vibrational analysis with first-principles calculations
- Identified the coupling of order parameters within the Landau Free Energy, suggesting that the material's properties are tunable via variation of applied strain or electric fields
- Presented initial findings and theoretical analysis in a poster presentation at the Lennard-Jones Centre, University of Cambridge

Nanjing University | Summer Research

July 2023 – September 2023

Project Topic:

Growth, critical property analysis, phase transition modulation, and ferroelectric domain studies of Perovskite single crystal

- Designed and built customized sample holders
- Conducted ferroelectric hysteresis tests across different temperatures to determine the Curie temperature and cryogenic behaviors of the perovskite structure, analyzing the results against theoretical predictions

SKILLS

Technical

First-principle calculation with VASP, Phonopy, Wannier90 and Quantum Espresso

Programming in Python, familiar with Matlab

Advanced Linux terminal operations and computing server management

Modeling with Auto desk and CAD, capable of fluid simulation with SimScale

Languages IELTS 8.0 in English (proficient); native Chinese speaker

Laboratory Transmission Electron Microscopy(TEM) operation, experiment planning, apparatus design

OTHER PROJECTS

Research Review: Unlocking the Potential of 2D Ferroelectrics with Machine Learning

- Reviewed recent advances in applying machine learning to 2D ferroelectrics, systematically organising methods and key findings.

Lennard-Jones Centre showcase poster presentation

- Shared findings from ferroelectric research project to an international audience with students and researchers from different fields of study

A-level Extended Project in fluid dynamic properties of paper air planes

- Planned and made literature review, experiment and data analyzing myself, got familiar with the basic process of a project
- Designed experiment which use a paper plane launcher and made the launcher from scratch
- Modeled with CAD and carried out fluid simulations

CERTIFICATION & AWARDS

2024 Clare College undergraduate research grant

2022 British Physics Olympiad (BPhO) Top Gold Award, BMO Qualification

2021 BPhO Top Gold Award, UKMT SMC Gold Award

EXTRACURRICULAR ACTIVITY

Former offensive lineman, Cambridge University Pythons American football club